

Project Title

Automated Student–Laboratory Allocation System Using Maximum Bipartite Matching

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

Laboratory allocation is often a challenging scheduling problem when multiple students prefer the same laboratory, and the available laboratory capacity is limited. The problem is that many students may prefer the same laboratory, but each laboratory has limited capacity.

The proposed system automates the allocation process using Maximum Bipartite Matching. Students and laboratories are represented as two separate sets of vertices in a bipartite graph. An edge is created between a student and a laboratory when the student prefers that laboratory.

The matching algorithm identifies an optimal set of students–laboratory assignments while ensuring that each student receives at most one laboratory. The system also generates an allocation report showing successful assignments and unallocated students.